

CHE 396 Quiz 4

Adisa Hadzic

TOTAL POINTS

77 / 100

QUESTION 1

11 77 / 100

- **0 pts** Correct

✓ - **5 pts** capital cost is wrong (only ISBL is taken into account)

✓ - **3 pts** Gross profit for first year of operation is wrong

✓ - **10 pts** Depreciation is wrong

- **2 pts** tax should be next year

- **5 pts** Discount factor is not right

- **0 pts** Click here to replace this description.

✓ - **5 pts** Gross profit is wrong

- **5 pts** tax is wrong

- **10 pts** cash flow is wrong

- **10 pts** taxable income is wrong

- **10 pts** PV is wrong or not done

- **10 pts** NPV is not calculated or wrong

- **2 pts** Depreciation is only taken for ISBL not full

capital cost

- **100 pts** Not done or wrong

ADISA HADZIL

Name

1. If ISBL cost of a plant is \$60,000 and annual fixed cost of production is \$15,000 and variable cost of production is \$10,000 and annual main product revenue is \$60,000, calculate the maximum 7 year NPV that can be achieved with a 15% interest rate and a 25% tax rate. Assume plant build in two years (capital paid in two years equally), and 50% operation the first year when operation starts. Use MACRS depreciation with a five-year recovery term. (MACRS depreciation charges for 5 year recovery, year 1 = 20, year 2 = 32, year 3 = 19.2, year 4 = 11.52, year 5 = 11.52 and year 6 = 5.76)

BEFORE TAX CF

YEAR	Gross profit	Depreciation charges	Taxable income	Taxes paid	Cash flow	Discount factor	Present value of CF	NPV
0	-60 000	0	0	0	-60 000	1	$= -60 000 \times 1 = -60 000$	-60 000
1	-25 000	0	0	0	-25 000	$\frac{1}{(1+0.15)} = 0.8696$	-21737.5	-81737.5
2	$0.50 \times 60 000 = 30 000$	$= 30 000 \times 0.2 = 6000$	24 000	0	30 000	0.7561	22683	-59054.5
3	60 000	$= 60 000 \times 0.32 = 19200$	40 800	6000	54 000	0.6575	35505	-23549.5
4	60 000	11520	48480	10200	49800	0.5718	28475.6	4926.1
5	60 000	6912	53088	12120	47880	0.4972	23805.9	28732
6	60 000	6912	53088	13272	46728	0.4323	20200.5	48932.5
7	60 000	3456	56544	13272	46728	0.3759	17565.1	66497.6

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$$\text{ISBL} = 60 000$$

$$\text{FCOP} = 15 000$$

$$\text{VCOP} = 10 000$$

$$\text{CCOP} = 25 000$$

$$\text{REVENUE} = 60 000$$

$$\text{CAPITAL COST} = \text{CCOP} + \text{ISBL} = 82 500$$

$$\text{ANNUAL REVENUE} = 60 000$$

$$\text{GP} = 60 000$$

MACRS

20

32

19.2

11.52

11.52

5.76

$$\text{DEPREC} = \text{GP} \times \text{MACRS \%}$$

$$\text{TAXABLE INC} = \text{GP} - \text{DEPREC}$$

$$\text{TAXES DUE} = \text{T. INC} \times 0.25$$

$$\text{C.F.} = \text{GP} - \text{TAX DUE}$$

$$\text{DP} = \frac{N}{(1+i)^n} \quad \text{PV.CF} = \text{DP} \times \text{CF}$$

NPV =

PV of CF + NPV (year before)